



Faculty of Engineering
Department of Biomedical Engineering

**UN Sustainable Development Affirmation for The Non-Contact
Clinical Thermometer**

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CHANGE RECORDS

FIRST RELEASE

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REVISIONS

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1. PURPOSE

The purpose of this affirmation is to demonstrate how the non-contact clinical thermometer project contributes to sustainable development, especially its alignment with the United Nations Sustainable Development Goals (SDGs). The project aims to support sustainable healthcare by improving hygienic clinical practices, enhancing patient safety, and promoting responsible use of medical resources through innovative, non-contact temperature measurement technology.

2. SCOPE

The scope of this affirmation covers the contribution of the non-contact clinical thermometer to sustainable development within healthcare systems. It focuses on the device's role in supporting public health, reducing infection risks, minimizing medical waste, and encouraging sustainable medical device design. The affirmation applies to the academic design and development phase and does not include large-scale manufacturing or commercialization processes.

3. SUSTAINABLE DEVELOPMENT

The sustainable development section is expressed in two-fold. First, the alignment of the project with sustainable engineering practices is analysed. This explores how the project embodies the core philosophy of sustainable engineering by being extremely reusable. Next, the alignment of the project with the UN Sustainable Development Goals(SDGs) is analysed. Finally, its commitment to sustainable and ethical design is explored.

Alignment With Sustainable Engineering Practices

I. Design For Disassembly

Our device is highly repairable. Inculcating the use of headers and other connectors ensures their configurations are reusable in multiple devices. This extends the product lifecycle, keeping a majority of the material in use, should any part get damaged.

II. Resource Efficiency

Our device was created with the mindset of maximizing our available resources. We prioritized selecting specific modules and subcomponents to perform the functions they are required. We also optimized our device by providing only the relevant materials, preventing excessive components that could be described as wastage.

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III. Economic Sustainability

Our device was created using COTS, which stands for Commercial-Off-The-Shelf components. These components are highly accessible on hobbyist platforms to encourage the replicability of the device. Our casing is also created using easily printable 3D designs.

IV. Environmental Sustainability

Since most parts of our device are highly accessible, creating our device required no imports. We hit our target of creating an environmentally friendly device.

V. Social Sustainability

Since this project was created by a team of highly cooperative individuals, every aspect of it was carried out with multiple contributors. This prevents intellectual waste as the knowledge of this process is being shared.

Alignment with the United Nations Sustainable Development Goals (SDGs)

This project supports the United Nations Sustainable Development Goals by contributing to sustainable healthcare practices through the ethical design of a non-contact clinical thermometer. The device aligns with the following SDGs:

SDG 3: Good Health and Well-Being



The non-contact clinical thermometer supports SDG 3 by enabling safe, hygienic, and rapid body temperature measurement in healthcare environments. Accurate temperature monitoring is essential for early detection of fever and effective patient assessment. The non-contact design reduces the risk of cross-infection, contributing to improved patient safety and overall public health outcomes.

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SDG 9: Industry, Innovation, and Infrastructure



The project promotes innovation in medical device design by integrating sensor technology and signal processing techniques to improve clinical measurement systems. The development of reliable, non-contact diagnostic tools contributes to strengthening healthcare infrastructure and encouraging technological advancement in the medical device industry.

SDG 12: Responsible Consumption and Production



The non-contact thermometer is designed with durability and efficient energy consumption in mind, reducing the need for disposable accessories and frequent replacements. This supports responsible use of resources and minimizes medical waste commonly associated with contact-based thermometers.

SDG 13: Climate Action (Indirect Contribution)



By minimizing disposable components and promoting reusable medical equipment, the device indirectly contributes to reducing healthcare-related waste and environmental impact, supporting broader climate action goals.

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Commitment to Sustainable and Ethical Design

The project affirms a commitment to developing medical technologies that improve clinical efficiency while adhering to ethical standards, environmental responsibility, and global sustainability objectives. The design prioritizes patient safety, hygiene, and accuracy, supporting long-term sustainable healthcare practices.

REFERENCES

- 3.1. United Nations, *Transforming Our World: The 2030 Agenda for Sustainable Development*.
- 3.2. United Nations, *Sustainable Development Goal 3: Good Health and Well-Being*.
- 3.3. World Health Organization (WHO), *Medical Device Design and Sustainable Healthcare*.
- 3.4. ISO 14971, *Medical Devices — Application of Risk Management*.

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